

General Exam

January 18th, 2023
(Time allowed: 90 minutes)

Notation: Recall that $\mathbf{1}(a < x < b) = \mathbf{1}_{(a,b)}(x)$ and that $f_X(x; \theta) = f_\theta^X(x)$.

Question 1 (7 points)

For $i = 1, \dots, n$, let X_i be i.i.d. according to the density function

$$f_X(x; \theta) = \frac{\theta 2^\theta}{x^{\theta+1}} \mathbf{1}_{[2, +\infty)}(x),$$

where $\theta > 0$ is an unknown parameter.

- (a) Find a complete sufficient statistic S for θ .

Solution: It is easy to show that the distribution belongs to the (full rank) exponential family. Hence $S = \sum \ln X_i$.

- (b) Find the MLE $\hat{\theta}$ of θ .

Solution: Since for $x_{(1)} \geq 2$

$$L(\mathbf{x}; \theta) = \frac{2^{n\theta} \theta^n}{(\prod x_i)^{\theta+1}},$$

we have

$$\ell(\theta) = n\theta \ln 2 + n \ln \theta - (\theta + 1) \sum \ln x_i, \quad \ell'(\theta) = n \ln 2 + \frac{n}{\theta} - \sum \ln x_i,$$

and we get a maximum if $\theta = \frac{n}{\sum (\ln x_i) - n \ln 2}$. Also, $\ell''(\theta) = -\frac{n}{\theta^2} < 0 \quad \forall \theta > 0$. Therefore

$$\hat{\theta} = \frac{n}{\sum (\ln X_i) - n \ln 2} = \frac{n}{\sum \ln \frac{X_i}{2}}.$$

- (c) Is $\hat{\theta}$ unbiased? Also evaluate the MSE of $\hat{\theta}$. [Hint: find the distribution of $Y = \ln \frac{X}{2}$]

Solution: It can be easily shown that $Y \sim \text{Gamma}(1, \theta)$, so that $\sum \ln \frac{X_i}{2} \sim \text{Gamma}(n, \theta)$. We have

$$\mathbb{E}(\hat{\theta}) = \frac{n}{n-1} \theta$$

and therefore $\hat{\theta}$ is biased. In order to evaluate the MSE, we first compute

$$\mathbb{E}(\hat{\theta}^2) = \frac{n^2}{(n-1)(n-2)} \theta^2,$$

from which we obtain $\text{Var}(\hat{\theta}) = \frac{n^2}{(n-1)^2(n-2)} \theta^2$. Consequently, we get

$$\text{MSE}_{\hat{\theta}}(\theta) = \frac{n+2}{(n-1)(n-2)} \theta^2.$$

- (d) First verify that the median of X is $x_m = 2^{1+\frac{1}{\theta}}$. Then find a $1-\alpha$ asymptotic confidence interval for $\tau(\theta) = x_m$. [Hint: use the invariance property of the MLE]

Solution: x_m satisfies $F_X(x_m; \theta) = 1/2$: since $F_X(x; \theta) = 1 - (2/x)^\theta$ for $x \geq 2$, we have

$$1/2 = F_X(x_m; \theta) = 1 - (2/x_m)^\theta \Rightarrow x_m = F_X^{-1}(1/2; \theta) = 2^{1+\frac{1}{\theta}}$$

If n is large enough, $\sqrt{n}(\hat{\theta}_n - \theta) \approx \mathcal{N}(0, \theta^2)$. Now, the invariance property of the MLE yields

$$\sqrt{n}(\tau(\hat{\theta}_n) - x_m) \approx \mathcal{N}\left(0, \frac{1}{\theta^2} [2^{1+\frac{1}{\theta}} \ln 2]^2\right).$$

Thus we obtain

$$x_m \in \left(\tau(\hat{\theta}_{L,n}) \pm z_{\frac{\alpha}{2}} \frac{2^{1+\frac{1}{\hat{\theta}_{L,n}}} \ln 2}{\hat{\theta}_{L,n} \sqrt{n}} \right).$$

Question 2 (9 points)

For $i = 1, \dots, n$, let X_i be i.i.d. according to the density function

$$f_X(x; \theta) = \frac{2x}{\theta^2} \mathbf{1}_{[0, \theta]}(x),$$

where $\theta > 0$ is an unknown parameter.

- (a) Find a minimal sufficient statistic S for θ .

Solution: By applying the Lehmann-Scheffé Theorem, one immediately obtains $S = \max X_i$.

- (b) Is S also complete?

Solution: First one needs to obtain the density of S , which turns out to be

$$f_S(s) = \frac{2ns^{2n-1}}{\theta^{2n}} \mathbf{1}_{[0, \theta]}(s).$$

Therefore, for any $\theta > 0$

$$\mathbb{E}(g(S)) = \int_0^\theta g(s) \frac{2ns^{2n-1}}{\theta^{2n}} ds = 0.$$

which is equivalent to

$$\int_0^\theta g(s) s^{2n-1} ds = 0.$$

Using Leibnitz's rule

$$g(\theta) \theta^{2n-1} = 0,$$

hence $g(s) = 0$ for $s > 0$ and S is complete.

- (c) Find $\hat{\theta}_{MM}$, the Methods of Moments estimator of θ .

Solution: We have $\mathbb{E}(X) = \frac{2}{3}\theta$. Hence $\hat{\theta}_{MM} = \frac{3}{2}\bar{X}$.

- (d) Find a $1 - \alpha$ confidence interval for θ based on $\hat{\theta}_{MM}$.

Solution: One has $\mathbb{E}(\hat{\theta}_{MM}) = \theta$ and immediately obtains $\text{Var}(X) = \frac{\theta^2}{18}$. By the CLT one has

$$\sqrt{n} \left(\bar{X}_n - \frac{2}{3}\theta \right) \approx \text{Norm} \left(0, \frac{\theta^2}{18} \right).$$

Therefore, applying the Delta Method, we obtain

$$\sqrt{n} (\hat{\theta}_{MM} - \theta) \approx \text{Norm} \left(0, \frac{\theta^2}{8} \right).$$

$$\theta \in \left(\hat{\theta}_{MM} \pm z_{\frac{\alpha}{2}} \frac{\hat{\theta}_{MM}}{\sqrt{8n}} \right)$$

- (e) Find $\hat{\theta}$, the MLE of θ . Is it unbiased?

Solution: Given

$$L(\mathbf{x}; \theta) = \frac{2^n \prod x_i}{\theta^{2n}} \mathbf{1}_{[\max x_i, +\infty)}(\theta),$$

we can see that the likelihood function is decreasing, so we obtain $\hat{\theta} = \max X_i$. Since $\hat{\theta} = S$, we have

$$\mathbb{E}(\hat{\theta}) = \mathbb{E}(S) = \int_0^\theta s \frac{2ns^{2n-1}}{\theta^{2n}} ds = \frac{2n}{2n+1} \theta.$$

Hence $\hat{\theta}$ is not unbiased, albeit it is asymptotically unbiased.

- (f) Find the UMVUE for θ .

Solution: Since S is a complete sufficient statistic for θ , by the Lehmann-Scheffé Theorem

$$T = \frac{2n+1}{2n} S = \frac{2n+1}{2n} \max X_i$$

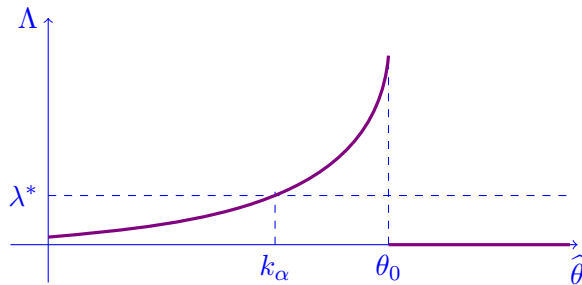
is the UMVUE.

- (g) Find the implementable form of the level α LRT to verify $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$.

Solution: The likelihood ratio is

$$\Lambda = \frac{L(\mathbf{x}; \theta_0)}{L(\mathbf{x}; \hat{\theta})} = \frac{\hat{\theta}^{2n} \mathbf{I}_{[\max x_i, +\infty)}(\theta_0)}{\theta_0^{2n} \mathbf{I}_{[\max x_i, +\infty)}(\hat{\theta})} \stackrel{\hat{\theta} = \max x_i}{=} \left(\frac{\hat{\theta}}{\theta_0} \right)^{2n} \mathbf{I}_{(0, \theta_0]}(\hat{\theta}).$$

We reject H_0 if $\Lambda < \lambda^*$:



then

$$\hat{\theta} < k_\alpha \quad \text{or} \quad \hat{\theta} > \theta_0,$$

with k_α such that $\mathbb{P}_{H_0}(\hat{\theta} < k_\alpha) = \alpha$. Since

$$\mathbb{P}_{H_0}(\hat{\theta} < k_\alpha) = \mathbb{P}_{H_0}(S < k_\alpha) = F_S(k_\alpha) = \left(\frac{k_\alpha}{\theta_0} \right)^{2n},$$

we have

$$k_\alpha = \theta_0 \sqrt[2n]{\alpha}.$$

(Note that the event $\hat{\theta} > \theta_0$ occurs with zero probability under H_0 .)

Question 3 (5 points)

Let X be a random variable having density function $f_X(x; \theta)$ with $\theta \in \mathbb{R}^1$.

- (a) Define

$$S_\theta(X) = \frac{\partial}{\partial \theta} \log f_\theta(X).$$

and show that $\mathbb{E}_\theta[S_\theta(X)] = 0$ for any $\theta \in \mathbb{R}^1$ (state also the necessary assumptions).

(b) Carefully define the Fisher Information $I_X(\theta)$.

Let (X_1, X_2, X_3) and $(Y_1, \dots, Y_8)$ be *i.i.d.* samples from $f_X(x; \theta)$. Prove or disprove the inequality

$$I_{(X_1, X_2, X_3)}(\theta) > I_{(Y_1, \dots, Y_8)}(\theta) \quad \text{for any } \theta \in \mathbb{R}^1.$$

Solution: Please refer to your notes and to the textbook.

Question 4 (5 points)

Carefully define sufficient, ancillary and complete statistics. State and prove Basu's theorem.

Solution: Please refer to your notes and to the textbook.

Question 5 (5 points)

State and prove in detail the Rao-Blackwell theorem. Explain why the UMVUE must be a function of the minimal sufficient statistic.

Solution: Please refer to your notes and to the textbook.